

BLOCKCHAIN LAB

Experiment 4

Title of Experiment:

"Creation of account in Metamask, Transfer of Amount/Cryptocurrency from one account to another."

1. Aim

To create a cryptocurrency wallet account and to transfer an amount/cryptocurrency from one account to another using a blockchain-based wallet, and to study and verify the resulting transaction details on the blockchain.

2. Theory

A cryptocurrency wallet is a software or hardware application that allows a user to store, manage, send, and receive digital assets on a blockchain network. Every wallet is associated with a public address (used to receive funds) and a private key (used to authorize and sign outgoing transactions). The private key must always be kept secret, since anyone who possesses it gains full control over the funds held in that wallet.

Wallets can broadly be classified as hot wallets and cold (hardware) wallets. A hot wallet, such as MetaMask, is a browser-based or mobile software wallet that remains connected to the internet, making it convenient for quick transactions. A cold wallet, such as a Ledger device, is a hardware wallet that stores the private keys offline on a physical device, and every transaction must be physically confirmed on the device before it is broadcast to the network. This makes hardware wallets significantly more secure against online attacks, since the private key never leaves the device.

For this experiment, a Ledger hardware wallet has been used instead of MetaMask. Functionally, both wallets operate on the same underlying principle: they interact with the Ethereum blockchain to create/manage an account, hold a public-private key pair, and broadcast signed transactions to the network. The steps involved — account creation, obtaining a public address, initiating a transfer, and confirming the transaction — remain conceptually identical in both wallets, the only difference being that MetaMask signs the transaction in software, whereas the Ledger device signs it in hardware for added security.

When a transaction is broadcast, it is picked up by miners/validators on the network, included in a block, and once that block is added to the chain the transaction is marked as "Confirmed." Every transaction has a unique Transaction ID (hash), a sender address (From), a receiver address (To), the amount transferred, and a network fee (gas fee) paid to the network for processing the transaction.

3. Steps / Procedure

1. Install the wallet application (Ledger Live for a Ledger hardware wallet, or the MetaMask browser extension if using MetaMask) and set it up on the system.
2. Create a new wallet account. This generates a public address and a private key/recovery (seed) phrase for the account. Note down and securely store the 24-word (or 12-word) recovery phrase — this is the only way to restore the wallet if the device/software is lost.
3. Connect the wallet to the desired blockchain network (e.g., Ethereum Mainnet) and verify the account balance.

4. To fund the wallet, obtain some test/actual cryptocurrency and note the public receiving address of the wallet.
5. Open the "Send" option in the wallet, enter the recipient's public address, and specify the amount/cryptocurrency to be transferred (in this experiment, 30 USDC was sent).
6. Review the transaction details, including the estimated network (gas) fee, and confirm/sign the transaction. On a Ledger device, this confirmation is physically approved on the hardware device itself.
7. Once signed, the transaction is broadcast to the blockchain network and awaits confirmation by miners/validators.
8. After the transaction is mined and added to a block, its status changes to "Confirmed," along with a block number/confirmation count.
9. Open "Transaction details" from the wallet's activity/history tab to view the complete record of the transaction — amount, date/time, network fee, Transaction ID, and the From/To addresses.
10. Verify the same transaction from the receiving account, where it will appear as a "Received" transaction with the corresponding equivalent amount credited.

4. Output

A wallet account was successfully created and 30 USDC was transferred from the sender's account to the receiver's account. The screenshots below (captured from the Ledger Live wallet) show the transaction details on both ends of the transfer.

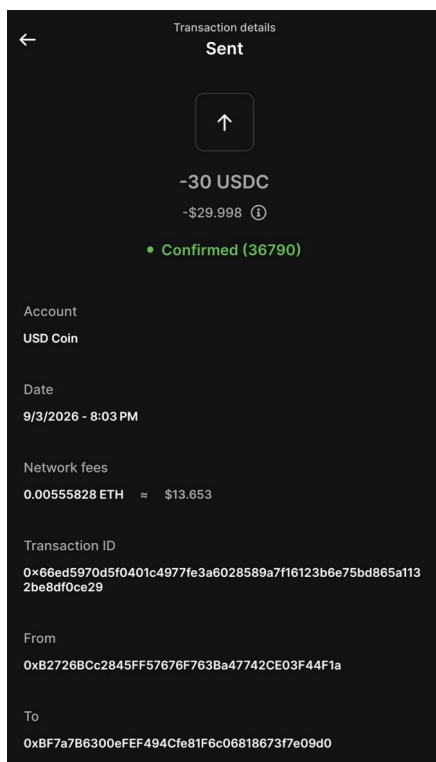


Fig. 1: Transaction details – Sent (Ledger)

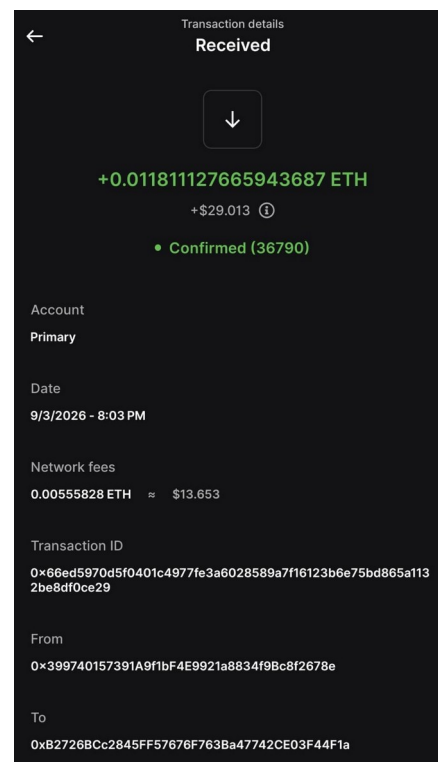


Fig. 2: Transaction details – Received (Ledger)

Observations from the screenshots:

- Transaction type: Sent (from the sender's USD Coin account) and Received (into the recipient's Primary/ETH account).

- Amount transferred: 30 USDC ($\approx$ $-\$29.998$) debited from the sender; an equivalent value of ≈ 0.011811 ETH ($\approx \$29.013$) credited on the receiving side.
- Date & Time: 9/3/2026, 8:03 PM.
- Network fee: 0.00555828 ETH ($\approx \$13.653$), paid to process the transaction on the network.
- Status: Confirmed, with confirmation/block number 36790.
- Transaction ID (hash): 0x66ed5970d5f0401c4977fe3a6028589a7f16123b6e75bd865a1132be8df0ce29 — this uniquely identifies the transaction on the blockchain and can be looked up on a block explorer.
- From address: 0xB2726BCc2845FF57676F763Ba47742CE03F44F1a | To address: 0xBF7a7B6300eFEF494Cfe81F6c06818673f7e09d0.

5. Conclusion

A cryptocurrency wallet account was successfully created and used to transfer 30 USDC from one account to another over the blockchain network. Although a Ledger hardware wallet was used in this experiment instead of MetaMask, the underlying process was found to be identical — both wallets generate a public–private key pair, allow the user to initiate and sign a transaction, broadcast it to the network, and display the confirmed transaction along with its Transaction ID, network fee, and sender/receiver addresses. This experiment thus provided practical, hands-on understanding of how account creation and peer-to-peer cryptocurrency transfers work on a blockchain, as well as the role of confirmations and network (gas) fees in validating transactions.